

# Chapter 3: Electric Resistance and Ohm's Law

## Formula Sheet + Tips & Tricks

### Formula Sheet

#### I. Electric Current and Current Density

##### Current and charge

$$Q = It$$

basic definition of steady current

##### Number of electrons

$$n = \frac{Q}{e}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

##### Variable current

$$Q = \int_{t_1}^{t_2} i \, dt$$

equals the area under the  $I$ - $t$  graph

#### II. Drift Velocity and Mobility

##### Microscopic form of current

$$I = nAev_d \Rightarrow v_d = \frac{I}{nAe}$$

$n$  = free-electron density

#### III. Resistance and Resistivity

##### Ohm's Law

$$V = IR \Rightarrow R = \frac{V}{I}$$

defines resistance  $R$

##### Resistance and resistivity

$$R = \rho \frac{L}{A} \Rightarrow \rho = \frac{RA}{L}$$

$$\text{conductivity } \sigma = \frac{1}{\rho}$$

#### Current density form

$$J = \sigma E = \frac{E}{\rho}$$

$$\rho = \frac{E}{J}$$

### IV. Change in Resistance on Stretching the Wire

#### Stretching a wire

$$A_1 L_1 = A_2 L_2$$

if length becomes  $n$  times, new resistance  $R' = n^2 R$

### V. Effect of Temperature on Resistance and Resistivity

#### Temperature dependence of resistance

$$R_T = R_0(1 + \alpha T)$$

$$\alpha = \frac{R_T - R_0}{R_0 T} = \text{temperature coefficient of resistance}$$

### VI. Combination of Resistances

#### Resistors in series

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$$

current is the same through each resistor

#### Resistors in parallel

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

voltage is the same across each resistor

### VII. Electric Power and Energy

#### Electric power

$$P = VI = I^2 R = \frac{V^2}{R}$$

electrical energy  $E = Pt$

## VIII. Electric Cell

## Terminal potential difference

$$V = \varepsilon - Ir \Rightarrow r = \frac{\varepsilon - V}{I}$$

while charging:  $V = \varepsilon + Ir$

## IX. Combination of Cells

## Cells in series

$$\varepsilon_{\text{total}} = \sum \varepsilon_i, \quad r_{\text{total}} = \sum r_i$$

for  $n$  cells connected in series

## Cells in parallel

$$\varepsilon_{\text{eff}} = \varepsilon, \quad r_{\text{eff}} = \frac{r}{n}$$

for  $n$  identical cells connected in parallel

## X. Kirchhoff's Laws

## Kirchhoff's Current Law

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

junction rule

## Kirchhoff's Voltage Law

$$\sum \varepsilon = \sum IR$$

loop rule, around any closed loop

## XI. Wheatstone Bridge and Metre Bridge

## Balanced Wheatstone bridge

$$\frac{P}{Q} = \frac{R}{S}$$

no current flows through the galvanometer

## Metre bridge

$$\frac{R}{S} = \frac{l}{100 - l}$$

$l$  = balance length measured from the left end

## XII. Potentiometer

## Potential gradient

$$k = \frac{\varepsilon}{R_w + r} \times \frac{R_w}{L}$$

potential drop per unit length (V/m)

## Potentiometer balance condition

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$$

emf is proportional to balance length

## XIII. Miscellaneous

## Circuit problem-solving

Ohm's Law + series-parallel reduction + Kirchhoff's Laws  
apply in combination, as per the circuit topology

## Tips, Tricks &amp; Hacks

- Drift velocity is tiny ( $\sim 10^{-4}$  m/s) — if an answer for  $v_d$  comes out large, you've likely mixed up  $n$  (electron density,  $\sim 10^{28}$  m $^{-3}$ ) with something else.
- Colour-code the two big “series vs parallel” rules: series adds resistances and keeps current the same; parallel adds conductances (reciprocals) and keeps voltage the same.
- Wheatstone bridge is balanced ( $\frac{P}{Q} = \frac{R}{S}$ )  $\Rightarrow$  galvanometer carries no current  $\Rightarrow$  you can just delete that branch when redrawing the circuit.
- Metre bridge is a Wheatstone bridge in disguise:  $\frac{R}{S} = \frac{l}{100 - l}$  — always measure  $l$  from the same end as  $R$ .
- In a potentiometer, the balancing length  $l$  is directly proportional to emf *only when no current is drawn* from the cell being measured — that's the whole point of using a potentiometer over a voltmeter.
- For power problems, choose the right form: if  $R$  is fixed and  $V$  varies, use  $P = V^2/R$ ; if  $I$  is fixed, use  $P = I^2 R$  — picking the wrong one wastes time squaring the wrong variable.
- Stretching a wire (volume constant): if length becomes  $n\times$ , resistance becomes  $n^2\times$  (since  $R \propto L^2$  when  $V = AL$  is fixed) — a guaranteed 1-mark question.
- In cell combinations, series is best when internal resistance  $\ll$  external resistance;

parallel is best when internal resistance  $\gg$  external resistance — know why, it's a common conceptual question.